

Add a query variable

Query variables allow you to write a data source query that can return a list of metric names, tag values, or keys. For example, a query variable might return a list of server names, sensor IDs, or data centers. The variable values change as they dynamically fetch options with a data source query.

Query expressions can contain references to other variables and in effect create linked variables. Grafana detects this and automatically refreshes a variable when one of its linked variables change.

Query expressions

Query expressions are different for each data source. For more information, refer to the documentation for your [data source](#).

Enter General options 🔗

1. Navigate to the dashboard you want to make a variable for and then click the **Dashboard settings** (gear) icon at the top of the page.
2. On the Variables tab, click **New**.
3. Enter a **Name** for your variable.
4. In the **Type** list, select **Query**.
5. (optional) In **Label**, enter the display name of the variable dropdown. If you don't enter a display name, then the dropdown label will be the variable name.
6. Choose a **Hide** option:
 - **No selection (blank)** - The variable dropdown displays the variable **Name** or **Label** value. This is the default.

- **Label** - The variable dropdown only displays the selected variable value and a down arrow.
- **Variable** - No variable dropdown is displayed on the dashboard.

Enter Query Options

1. In the **Data source** list, select the target data source for the query. For more information about data sources, refer to [Add a data source](#).
2. In the **Refresh** list, select when the variable should update options.
 - **On Dashboard Load** - Queries the data source every time the dashboard loads. This slows down dashboard loading, because the variable query needs to be completed before dashboard can be initialized.
 - **On Time Range Change** - Queries the data source when the dashboard time range changes. Only use this option if your variable options query contains a time range filter or is dependent on the dashboard time range.
3. In the **Query** field, enter a query.
 - The query field varies according to your data source. Some data sources have custom query editors.
 - If you need more room in a single input field query editor, then hover your cursor over the lines in the lower right corner of the field and drag downward to expand.
4. (optional) In the **Regex** field, type a regex expression to filter or capture specific parts of the names returned by your data source query. To see examples, refer to [Filter variables with regex](#).
5. In the **Sort** list, select the sort order for values to be displayed in the dropdown list. The default option, **Disabled**, means that the order of options returned by your data source query will be used.
6. (optional) Enter [Selection Options](#).
7. In **Preview of values**, Grafana displays a list of the current variable values. Review them to ensure they match what you expect.
8. Click **Add** to add the variable to the dashboard.

Related Grafana resources

